Question 4b: The Simple Version

You've Got This!

1 Don't Panic - Here's What's Really Being Asked

Question 4b sounds complicated, but it's actually asking something very intuitive:

> **The Real Question:**
> "I know there were n arrivals by time s. What can I say about arrivals at some OTHER time t?"
> That's it. That's the whole question.

The trick is: your answer depends on whether t is BEFORE or AFTER time s.

2 Let's Use a Concrete Story

Imagine customers arriving at a coffee shop. On average, $\lambda = 3$ customers per hour arrive.

The Fact You Know: By 9am (time $s = 9$), exactly 13 customers have arrived ($N_9 = 13$).

Now the question asks: *"What about arrivals at time t?"*

3 Case 1: Looking BACKWARD (When $t < s$)

3.1 The Question

You're at 9am. You know 13 customers have arrived by now.

Question: How many of those 13 customers had arrived by 4am?

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Time:    midnight    4am         9am
         |-----------|-----------|
                     ?           13 customers total
```

3.2 The Answer: Binomial

Think about it: You have 13 customers. Each one arrived at some random time between midnight and 9am.

For each customer:

- 4am is $\frac{4}{9}$ of the way from midnight to 9am
- So each customer has a $\frac{4}{9}$ chance of arriving before 4am

- Like flipping a coin with probability $\frac{4}{9}$ for each of the 13 customers

Answer for Case 1

$$(N_4 \mid N_9 = 13) \sim \text{Binomial}\left(13, \frac{4}{9}\right)$$

Translation: Out of 13 customers, each has a $\frac{4}{9}$ chance of being in the first 4 hours.
Mean: On average, $13 \times \frac{4}{9} \approx 5.78$ customers arrived by 4am.
Notice: We DON'T need to know $\lambda = 3$! The rate doesn't matter when looking backward.

3.3 Why This Makes Sense

You already know the 13 customers arrived somewhere between midnight and 9am. You're just asking: "How are they split between [midnight-4am] and [4am-9am]?"

Since arrivals are random, they split proportionally: $\frac{4}{9}$ vs $\frac{5}{9}$.

4 Case 2: Same Time (When $t = s$)

4.1 The Question

You're at 9am. You know 13 customers have arrived.

Question: How many customers arrived by 9am?

4.2 The Answer

Obviously 13. You just told me 13 customers arrived by 9am!

Answer for Case 2

$$N_9 = 13 \quad \text{(for certain)}$$

This case is trivial - if $t = s$, then $N_t = n$ with 100% certainty.

5 Case 3: Looking FORWARD (When $t > s$)

5.1 The Question

You're at 9am. You know 13 customers have arrived so far.

Question: How many customers will have arrived by 2pm (time $t = 14$)?

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Time:    midnight    9am         2pm
         |-----------|-----------|
         13 customers    ???
```

5.2 The Answer: n + Poisson

Think about it:

- You DEFINITELY have the 13 customers who already arrived
- PLUS however many MORE arrive between 9am and 2pm (5 hours)
- Those future arrivals follow a Poisson with rate $\lambda \times 5$ hours

Answer for Case 3

$$(N_{14} \mid N_9 = 13) = 13 + \text{Poisson}(\lambda \times 5) = 13 + \text{Poisson}(3 \times 5) = 13 + \text{Poisson}(15)$$

Translation: 13 known customers PLUS a Poisson(15) random number of future customers.
Mean: On average, $13 + 15 = 28$ customers by 2pm.
Notice: NOW we need $\lambda = 3$! We need to know the rate to predict the future.

5.3 Why This Makes Sense

You can't change the past - you know 13 customers already came. But the future is random - you expect about 15 more customers (since $\lambda = 3$ per hour $\times$ 5 hours = 15), but it could be more or less.

6 Side-by-Side Comparison

Aspect	**Case 1: $t < s$ (Past)**	**Case 3: $t > s$ (Future)**
Example	4am vs 9am	9am vs 2pm
Known info	13 by 9am	13 by 9am
Question	How many by 4am?	How many by 2pm?
Distribution	Binomial$(13, \frac{4}{9})$	13+ Poisson(15)
Need λ?	NO	YES
Mean	$13 \times \frac{4}{9} \approx 5.78$	$13 + 15 = 28$
Intuition	Split 13 proportionally	Known + future

7 The General Formula (What 4b Wants)

For ANY times t and s:

Complete Answer

Given $N_s = n$, the distribution of N_t is:

If $t < s$ (looking at the past):

$$N_t \sim \text{Binomial}\left(n, \frac{t}{s}\right)$$

If $t = s$ (same time):

$$N_t = n \quad \text{(certain)}$$

If $t > s$ (looking at the future):

$$N_t = n + \text{Poisson}(\lambda(t-s))$$

8 How to Actually Solve Problems

8.1 Step 1: Identify Which Case

Look at t and s. Is $t < s$, $t = s$, or $t > s$?

8.2 Step 2: Use the Right Formula

- Past ($t < s$): Binomial - use $\frac{t}{s}$ as the probability
- Same ($t = s$): Just n
- Future ($t > s$): n + Poisson - need λ

8.3 Step 3: Calculate

Plug in numbers and compute!

9 Practice Problem

Given: $\lambda = 2$ customers per hour. By time $s = 5$ hours, $n = 8$ customers arrived.

9.1 Question A: How many by time $t = 2$ hours?

Identify: $t = 2 < s = 5$ (PAST)
Formula: Binomial$(8, \frac{2}{5})$
Mean: $8 \times \frac{2}{5} = 3.2$ customers
Full answer: $(N_2 \mid N_5 = 8) \sim \text{Binomial}(8, 0.4)$

9.2 Question B: How many by time $t = 10$ hours?

Identify: $t = 10 > s = 5$ (FUTURE)

Formula: $8 + \text{Poisson}(\lambda(t - s)) = 8 + \text{Poisson}(2 \times 5) = 8 + \text{Poisson}(10)$

Mean: $8 + 10 = 18$ customers

Full answer: $(N_{10} \mid N_5 = 8) = 8 + \text{Poisson}(10)$

10 Common Mistakes (and How to Avoid Them)

10.1 Mistake 1: Using Binomial for Future

Wrong: "By 9am there were 13 customers. By 2pm: Binomial$(13, \frac{14}{9})$"

Why wrong: You can't have probability > 1! Also, you can't have fewer than 13 customers at 2pm.

Right: Use $13 + \text{Poisson}$ for future times.

10.2 Mistake 2: Forgetting λ for Future

Wrong: "I don't need λ for any case"

Why wrong: For the future, you MUST know the rate to predict new arrivals.

Right: Past doesn't need λ, but future does!

10.3 Mistake 3: Using Wrong Time Interval

Wrong: Future case: $n + \text{Poisson}(\lambda t)$

Why wrong: Should be $\text{Poisson}(\lambda(t - s))$ - the time from s to t, not from 0 to t.

Right: Use the DIFFERENCE: $t - s$

11 You've Got This!

The Big Picture:

- This is just asking: "Given info at time s, what about time t?"
- Past = split what you know (Binomial)
- Future = known + random new arrivals (Poisson)
- It's actually intuitive once you think about the coffee shop story!

How to succeed:

1. Always identify: is t before or after s?
2. Use the right formula
3. For past: just need n and $\frac{t}{s}$
4. For future: need n, λ, and $(t - s)$

You can do this. The concept is actually simpler than it looks. Just remember the coffee shop!